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Research article

GUIDE2FR: A smart monitoring platform with a digital twin of a firefighter training tower for emergency scenarios

Marcos Delgado Álvaro [a], Robert Novak [a], Pedro Rafael Fernández Barbosa [a], Iván Chicano Capelo [b], Micael Gallego [b], M. Cristina Rodriguez-Sanchez [a,*]

[a] Department of Electronic Technology, Rey Juan Carlos University, c/ Tulipán, s/n, Móstoles, 28933, Spain
[b] Department of Computer Science and Statistics, Rey Juan Carlos University, c/ Tulipán, s/n, Móstoles, 28933, Spain

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ABSTRACT

This paper describes the implementation of a digital twin for buildings to enhance the emergency response capabilities of first responder teams, including firefighters, police, and emergency medical services. The proposed platform improves the planning of preventive evacuation strategies and supports real-time operational decisions during emergencies. It integrates wireless monitoring beacons, specifically designed for hostile environments, a cloud-based data management system, and a predictive model to monitor environmental air quality parameters, which are critical during emergency scenarios. The platform uses a digital twin to simulate the building's behavior, incorporating multimedia content and time series graphs to enhance situational awareness and decision-making. Real-time building data are seamlessly integrated with predictive models generated from smart building sensors, offering a comprehensive visualization on a web-based monitoring interface. This approach provides critical insights to decision-makers, improving the safety and efficiency of both rescue operations and preventive measures.

1. Introduction

A Smart City is an urban area that uses advanced information and communication technologies (ICT) to improve the quality of life of its residents, optimize resource usage, improve the efficiency of urban services and promote sustainability. It integrates data from various sources, such as IoT (Internet of Things) devices and sensors, to enable real-time monitoring, predictive analytics, and informed decision-making. Additionally, Smart Cities foster citizen engagement and collaboration, ensuring that technological advancements align with the needs and well-being of the community. These technologies are integrated into key sectors, such as transportation, waste management, energy, security, and infrastructure, to create a more sustainable, efficient, and livable environment [1]. Smart Cities and smart buildings are closely related, sharing the common goal of improving efficiency, sustainability, and quality of life through advanced technologies. Smart buildings are critical components of Smart Cities, playing a pivotal role in enhancing efficiency, sustainability, and urban livability [2]. These structures leverage advanced technologies such as IoT, AI (Artificial Intelligence), and automation to monitor and optimize energy consumption, improve occupant comfort, and ensure operational efficiency. By incorporating real-time data analysis and adaptive management systems, they help reduce the environmental footprint of cities while fostering safety and convenience for their users.

* Corresponding author.
E-mail addresses: marcos.delgadoal@urjc.es (M.D. Álvaro), robert.novak@urjc.es (R. Novak), pedro.barbosa@urjc.es (P.R.F. Barbosa), ivan.chicano@urjc.es (I.C. Capelo), micael.gallego@urjc.es (M. Gallego), cristina.rodriguez.sanchez@urjc.es (M.C. Rodriguez-Sanchez).

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Smart buildings are fully integrated with urban infrastructure via communication networks and IoT-enabled sensors. These systems collect and share data on various operational aspects, including energy usage, climate control, and security, which is then aggregated into the city network. This integration optimizes resource allocation and enhances public services. For instance, building's energy consumption can be dynamically adjusted based on the city's overall demand, contributing to electrical grid stability and efficiency [3,4].

One key area of focus within Smart Cities is safety. Buildings with established safety, evacuation, and environmental monitoring plans contribute significantly to urban security [5]. This paper focuses on this crucial aspect, offering supportive tools aimed at providing robust, proactive solutions that directly influence the safety and planning efficiency of first responder teams operating in emergency or high-risk scenarios. These solutions facilitate early detection of incidents or risk assessment and monitoring, enabling a swift return to normalcy and efficient organization of security forces in the event of a disaster.

In this context, virtualization through digital twin technology has emerged as an effective tool for transforming urban emergency services, offering faster information and more efficient operations. Integrating monitoring systems to measure relevant environmental variables and asset management strategies are key aspects designed and evaluated in this work to enhance operational efficiency. Various studies have explored the effectiveness of monitoring in emergency situations [6–11].

According to the report on emergency services and fire-related fatalities in Spain, published by *Fundación Mapfre* in collaboration with the *Asociación Profesional de Técnicos de Bomberos* (APTB) [12], there were 235 fire and explosion-related fatalities in 2022. Of these, 220 were fire-related deaths, with 176 occurring in residential buildings. These fatalities are often caused by direct exposure to fire, smoke inhalation, burns, stress, or difficulty escaping enclosed spaces. Other common risks during emergencies include gas leaks, such as ammonia, or explosions. Therefore, action and evacuation plans must be highly accurate, emphasizing coordination and communication, along with real-time information about the incident.

In recent years, the concept of Smart Cities has expanded to incorporate more advanced strategies for urban safety and emergency response, particularly through the integration of digital twin technologies. These platforms enable synchronized multimodal emergency monitoring and real-time visualization of urban infrastructure. Recent studies have demonstrated how digital twins can be applied in civil and infrastructure emergency management [13] and in fire safety within smart buildings [14]. Moreover, real-time digital twins are gaining relevance as foundational technologies for the next generation of networked systems, such as 6G and beyond, enabling ultra-reliable low-latency communication essential in time-critical scenarios like emergency response [15]. These developments highlight the growing trend toward integrated, intelligent platforms for situational awareness and decision-making in public safety applications, underscoring the relevance of our proposed platform.

One of the main objectives of the system presented here is to provide accessible tools for intervention teams facing emergency scenarios, minimizing response times. This, in turn, reduces the risk of incidents or casualties, both among civilians and emergency responders.

Technologies supporting first responder teams have been explored and evaluated in this work, including solutions presented in [16,17], which align with our research focus. Both studies highlight the importance of leveraging technology to enhance environmental monitoring for strategic oversight, both in emergencies and during normal conditions. Other solutions, such as those described in [18], also utilize such resources to provide alternatives.

This work explores the impact of employing Building Information Modeling (BIM) methodologies to facilitate information access and operation visualization. BIM enables building virtualization, creating a comprehensive database of assets and installations, accessible and updated in real-time [19]. The creation of a building's digital twin involves establishing information exchange standards throughout its lifecycle, with ISO 19650 [20] being the relevant European standard. The Universidad Rey Juan Carlos (URJC) has experience in building digitization, both for its campuses and external entities, through collaborative projects [16]. This digitization process involves laser scanning and 3D modeling, providing detailed information on numerous objects and assets, including their material types, models, etc. This data can be further enhanced with information about the building's use and condition (occupancy, schedules, emergency exits, fire extinguishers, evacuation routes, emergency plans, etc.).

Beyond the digital twin, the integration of real-time data from physical variables, such as temperature or CO_2 levels in different building areas, is crucial for emergency management. Real-time visualization allows remote area monitoring and segmentation, supporting efficient operational responses. Achieving this requires a flexible software platform capable of processing data from multiple sources, capturing, storing, and displaying sensor and multimedia content (videos, photos). This approach would help better categorize emergencies and establish an environmental and visual media repository.

To implement this, IoT devices equipped with sensors and actuators create an interconnected network using various wireless communication technologies such as Wi-Fi, Bluetooth and LoRa. All this information must be efficiently transmitted, managed, and stored by the platform, ensuring rapid and intuitive user access through a data integration platform and a web interface that visualizes the digital twin, including graphs and other relevant information.

This work introduces GUIDE2FR, a platform designed for data acquisition and visualization, tested at the firefighter training tower located at the Unified Security Center (CUS) in Alcorcón (Madrid). Sensor beacons were deployed, and real emergency scenarios were simulated during firefighter training exercises, allowing the Advanced Command Post (PMA) to make informed decisions in line with emergency service protocols. These sensor beacons can be deployed autonomously, manually or pre-intervention as if they were the building's own monitoring systems. Flexibility and scalability are two of the great advantages of this system, as it adapts to existing and new information systems. Besides real-time visualization, the platform offers environmental parameter prediction capabilities, such as CO_2 levels, using machine learning techniques.

In summary, GUIDE2FR provides significant innovations compared to existing commercial solutions: full integration of real-time environmental and structural data, predictive capabilities through AI, flexible and scalable architecture, and synchronized

sensor data and video visualization. Being largely open-source, it offers greater customization and adaptability, making it a more comprehensive and effective tool for urban emergency management.

The rest of the document is organized as follows: Section 2 reviews Smart Building tools used for the proposed platform; Section 3 describes the architecture and data flow; Section 4 presents the results; and finally, Section 5 discusses the conclusions of the work.

2. Methodology for smart building solutions in emergency scenarios

The GUIDE2FR platform combines hardware and software elements to manage and display information. On the hardware side, the system includes sensors and gateways that support various communication protocols. The software tools are organized into multiple layers: a data integration layer and a visualization layer, using web tools to display the digital twin and related data. Each component is described in the following subsections.

2.1. Sensor layer — Beacon systems

It consists of a distributed system of beacons integrating smart IoT-based sensors. This setup provides data on various physical variables that need to be measured, such as temperature, humidity, CO_2, and total volatile organic compounds (TVOC), as well as other parameters depending on the building's usage and characteristics. This information can be obtained from various sources, such as sensors strategically placed in specific locations, and can utilize different wireless communication protocols. In this case, both *Wi-Fi* and *LoRa* communication technologies have been considered. Additionally, this array of sensors is complemented by strategically placed cameras used for video recording and monitoring critical areas. For instance, cameras with RTSP (*Real-Time Streaming Protocol* [21]) and *Wi-Fi* connectivity have been implemented in this work. RTSP is an application-level protocol used in surveillance systems and IP cameras to control real-time video and audio streaming. It allows communication between media servers and client devices. Several types of solutions will be validated in this study:

- Beacons comprising sensors will be pre-positioned across various spaces within the building.
- Beacons will be deployed by first responder teams at the onset of an emergency.
- Beacons will be autonomously distributed by ground or aerial autonomous agents (robots or UAVs) to minimize human intervention.

2.1.1. DBEMERGENCY beacons

During the proof of concept, the beacons were designed and implemented in collaboration with *Deep-Insight* to meet the specific requirements of the platform. These beacons are environmental monitoring devices optimized for emergency environments, featuring a modern IoT architecture. Under normal conditions, the beacons connect to the building's power supply and *Wi-Fi* network, transmitting air quality and environmental parameter data. In emergency situations, such as power or communication outages, they continue to operate using internal batteries and *LoRa* communication, facilitated by an EVA (a custom-designed device that serves as a remote access unit equipped with *LoRa* and Internet interface).

The development and implementation of platforms like GUIDE2FR are particularly crucial in environments where pre-existing connectivity is either insufficient or entirely unavailable. In many emergency or high-risk scenarios — such as firefighter training towers, remote industrial facilities, or large public buildings — the absence of reliable communication networks can significantly hinder monitoring and response efforts. Therefore, establishing a dedicated communication and monitoring network in these areas is vital for the safety and efficiency of emergency operations.

The beacon system in GUIDE2FR addresses this challenge by creating its own robust communication infrastructure. This is achieved through the integration of dual communication protocols: *Wi-Fi* and *LoRa*. In areas where traditional *Wi-Fi* networks are unavailable or compromised, the *LoRa* protocol ensures continuous, long-range, and low-power data transmission. This capability allows the beacons to maintain reliable connectivity across extensive areas or in buildings with structural barriers that could impede standard wireless signals.

By establishing a self-sustained network, the system ensures that crucial environmental data — such as temperature, humidity, and air quality parameters — are continuously monitored and transmitted. This information is essential for first responders and emergency teams, as it provides real-time insights into hazardous conditions and informs critical decision-making processes.

The main features of the *DBEMERGENCY* beacons, designed to support this self-sufficient monitoring network, are summarized in Table 1 and detailed below:

1. **Integrated sensors**: The beacons include sensors for temperature, humidity, atmospheric pressure, real CO_2, total volatile organic compounds (TVOC), air quality index (IAQ), motion detection, luminosity, and three-axis acceleration. The sensors provide essential measurements, with key ranges for human safety as follows:

    - **Temperature**: Typically considered safe for human occupancy between 20 °C and 27 °C. Values outside this range may pose risks such as heat stress or hypothermia.
    - CO_2: Normal indoor concentrations range from 400 to 1000 ppm. Levels above 1000 ppm can cause discomfort and impaired cognitive performance, while concentrations exceeding 5000 ppm are hazardous.
    - **TVOC**: Safe levels are generally below 500 ppb (parts per billion). Higher concentrations may lead to irritation or long-term health risks.

Table 1
Environmental data acquisition system components.

Component	Description
Sensor network	Distributed beacons equipped with environmental sensors (temperature, humidity, CO_2, etc.) capturing real-time data.
Primary communication protocol	*Wi-Fi* (IEEE 802.11 standards) used for initial data transmission.
Backup communication protocol	*LoRa* (LoRaWAN) serves as a fallback to ensure continuous data flow in the event of *Wi-Fi* failure.
Gateway	Aggregates sensor data and transmits it over the Internet to cloud storage, supporting both communication protocols.
Data management	*Python* scripts automate data retrieval from the cloud API, keeping the system updated for real-time analysis.

- **Humidity**: Optimal indoor humidity ranges from 30%–60%. Levels outside this range can affect respiratory comfort and increase the risk of mold growth.

2. **Connectivity and communications**: The beacons support *Wi-Fi* (IEEE 802.11 standards) and *LoRa* (LoRaWAN specifications). Dual transmission prioritizes *Wi-Fi* but switches to *LoRa* in emergencies to ensure continuous data flow.
3. **Configurable data transmission rates**: The transmission intervals can be set to 10 s, 30 s, or 60 s. Battery autonomy is adjusted accordingly: 2.5 days at a 10 s interval, 3.5 days at 30 s, and up to 5 days at a 60 s interval.
4. **Emergency operation**: Even without *Wi-Fi* or external power, the beacons continue to transmit data via *LoRa*, using the EVA unit as a network bridge. This ensures that the most recent and critical data is always available, enhancing situational awareness for emergency teams.
5. **Configuration aspects**: The beacons are wirelessly configured via *Wi-Fi*, and real-time data is accessible through the manufacturer's API, ensuring seamless integration with the monitoring platform.

2.1.2. Deployment in evaluation environments

Sensor beacons were strategically deployed in two key environments for evaluation: the firefighter training tower in Alcorcón and 10 classrooms distributed across Laboratory Building III at the Móstoles campus of URJC. The collected data is securely stored in an external database, enabling both real-time monitoring and comprehensive post-event analysis.

Environmental data acquisition occurs through the beacon network, capturing physical variables for storage and visualization. As previously mentioned, the beacons primarily use Wi-Fi, switching to LoRa as needed for robust communication. A gateway sends sensor data to the manufacturer's cloud storage, with an API for retrieving data. *Python* scripts are used to ensure data remains updated in the application.

2.2. Video management using OpenVidu

To manage video data captured by the cameras, a module was developed based on the open-source video conferencing platform *OpenVidu*.[1] *OpenVidu* is an open-source platform that facilitates the integration and implementation of real-time applications, particularly those requiring real-time video and audio streaming. The platform is built on *WebRTC*[2] technology and provides a high-level API, simplifying real-time video integration for web and mobile applications.

WebRTC is an open-source project that enables browsers and mobile applications to communicate in real time via simple APIs. It supports video, audio, and generic data transmission of any kind between peers, allowing developers to build robust communication solutions. WebRTC employs several underlying protocols and codecs to achieve this.

The most important protocols used for transporting media are:

- *Session Description Protocol (SDP)*: Used for negotiating the media parameters and capabilities. SDP is a format for describing the multimedia content of the communication sessions such as codec, format information, network information, and other media session details [22].
- *Real-time Transport Protocol (RTP)*: A network protocol used for delivering audio and video over IP networks [23].
- *Interactive Connectivity Establishment (ICE)*: A framework to allow WebRTC to connect peers through NATs and firewalls [24]. ICE uses STUN (Session Traversal Utilities for NAT) [25] and TURN (Traversal Using Relays around NAT) [26] protocols to find the best path to connect peers.

The codecs used are the following:

[1] https://openvidu.io/.
[2] https://webrtc.org/.

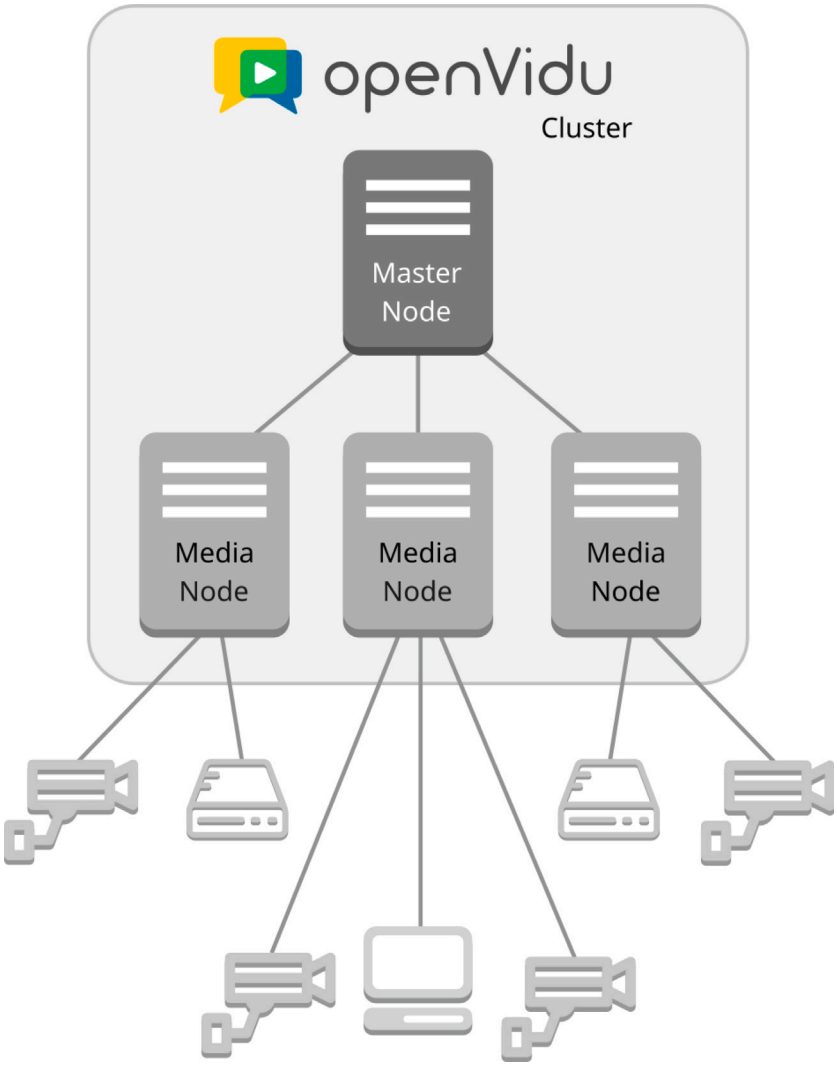


Fig. 1. Architecture of *OpenVidu*. The *OpenVidu* platform includes a master node and one or more media server nodes. Client devices, such as IP cameras, connect through the master node, which assigns them to a media server node.

- *VP8*: Widely used codec for web video compression. VP8 provides high-quality video at lower bit rates, typically around 500 kbps to 2 Mbps for standard definition (SD) video, and up to 4 Mbps for high definition (HD) video. It supports resolutions up to 4 K [27].
- *Opus*: Supports constant and variable bitrate encoding from 6 kbit/s to 510 kbit/s, frame sizes from 2.5 ms to 60 ms, and various sampling rates from 8 kHz (with 4 kHz bandwidth) to 48 kHz (with 20 kHz bandwidth, covering the entire human auditory range) [28].

OpenVidu also provides out-of-the-box features for support of IP cameras and video recording, allowing the platform to capture video streams from cameras and store them for later analysis on demand, simplifying the integration of video data into the monitoring platform and making it the best choice for our use case. The *OpenVidu* platform consists of the modules illustrated in Fig. 1:

- **Master Node**: Runs an application that controls one or more media nodes. It manages the control plane, handles sessions, forwards events and messages to clients, and distributes load among the available media nodes.
- **Media Node**: Runs the media server responsible for the media plane. The *OpenVidu* platform can use multiple media nodes, allowing for horizontal scaling. Each media node can handle multiple cameras, providing a scalable solution for any number of devices.

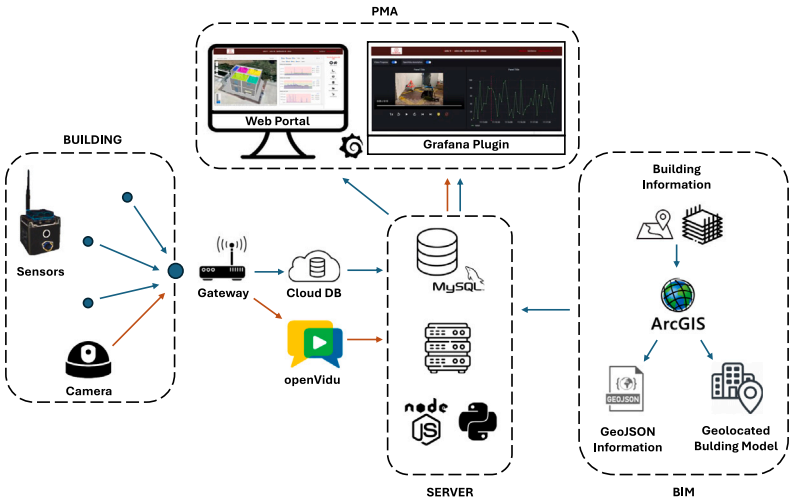


Fig. 2. Block diagram of the GUIDE2FR platform architecture.

Currently, the Media node utilizes *Kurento Media Server*,[3] an open-source implementation of a media server that supports the *WebRTC* standard and provides remote management of the *GStreamer* multimedia framework.[4] *Kurento* also works with RTP senders and receivers, enabling the platform to connect to IP cameras using RTSP. It also allows to consume the media streams received from the cameras via RTSP and store them into local or remote file systems, allowing *OpenVidu* to make use of these recordings as stated previously. Both *OpenVidu* and *Kurento* have been developed at Universidad Rey Juan Carlos.

In this work, cameras with *Wi-Fi* connectivity and RTSP were tested, similar to surveillance cameras used in buildings. The cameras connect to a *Kurento* media server through *OpenVidu*. The media server handles real-time video and audio streaming across the network. Video and audio captured by the cameras are stored as mp4 recordings in the server's file system by *OpenVidu*.

The data storage system is composed of a *MySQL* database, specifically structured to be scalable and support fast, flexible queries. This database is used to maintain a historical record of variables and states of interest. Video data obtained from the cameras is stored in the server's file system in several-second fragments. Additionally, the start date and time of each video fragment are stored in the database for efficient control and quick access to historical recordings.

2.3. Visualization layer — Digital twin

The visualization layer enables users to understand what is happening inside the building. It provides real-time information on current events, aiding in decision-making for evacuations or emergencies by visualizing rooms, corridors, and maintenance or furniture elements.

Additionally, it grants access to historical data, which is valuable for incident analysis or to create datasets for training models used in the data prediction platform. The visualization layer is implemented as a *web* environment using *Angular*, creating a portal that integrates a geolocated 3D model of the entire building along with historical graphs and sensor data. The 3D model is geolocated using *ArcGIS*, creating a contextual building scene. To visualize both historical and real-time data, this proposal uses *Grafana*, with *Fiware* serving as the data integration platform. The visualization layer synchronizes images from RTSP cameras in the building's key areas with sensor information, using a custom *Grafana* plugin that aligns the video player timeline with the graphical representation of data values. The aligning is done using the video and historical record data recorded from *OpenVidu* that resides in the server's file system and in the database respectively.

3. GUIDE2FR architecture description

The GUIDE2FR platform is designed to collect building data and display it through a *web* portal. The system manages various sensors, regardless of the technology used or the data's source and destination. Sensor information is transmitted to the manufacturer's cloud storage service. This system provides an API for data retrieval and forwarding. A *Python* application running on a server handles data management and redirects it to a *MySQL* database. Finally, the visualization component integrates sensor information, the building's 3D model, and *Grafana* dashboards for the end-user. Fig. 2 illustrates the block diagram of the different elements in the architecture, showing the data flow from the sensors to the *web* service. Additionally, the video management module,

[3] https://doc-kurento.readthedocs.io/en/latest/.
[4] https://gstreamer.freedesktop.org/.

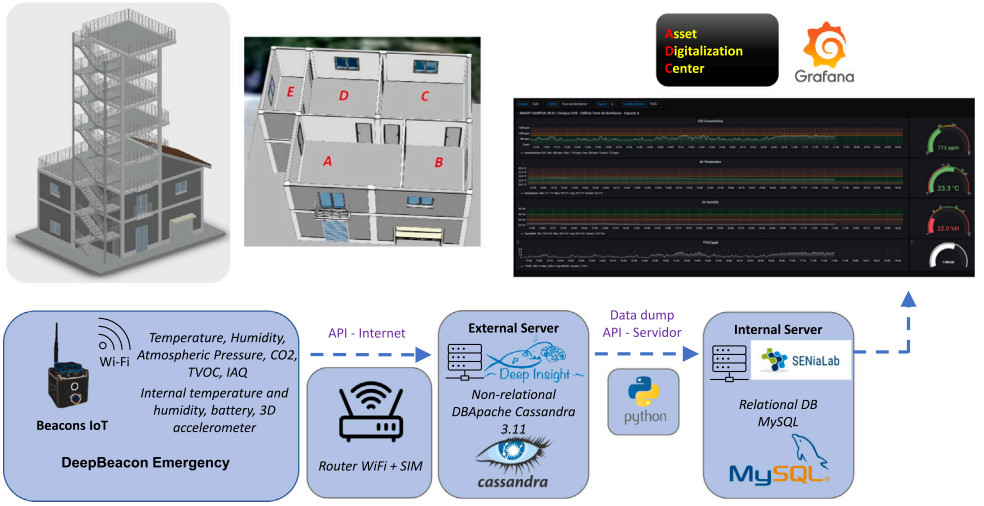


Fig. 3. Proposed architecture for sensor data transmission.

developed using *Node.JS*, operates on the same server as the main application responsible for requesting the capture of video footage to *OpenVidu* and storing said video footage on disk and saving the relevant metadata in *MySQL*. This metadata is used to create *Grafana* dashboards that synchronize time-series sensor data with video frames.

GUIDE2FR's architecture is designed with scalability in mind. Its architecture uses Docker containers for modular deployment of services like *Fiware*, *MySQL*, *OpenVidu*, Video Management Module, *Grafana*, Web Backend and the *Angular* frontend. These modules can be deployed in a scalable way to handle increased load. For example, *OpenVidu* can be horizontally scaled with several media nodes as shown in Figure 1 (elastic deployment), and load testing has been performed in *OpenVidu* platform with different media servers to know how many media streams is able to handle depending on hardware and media session topology [29]. The Video Management Module can be replicated and deployed in several servers to control as many RTSP cameras as needed. *MySQL* can be configured to be scalable leveraging tools like sharding, partitioning, and read replicas. The use of LoRaWAN and Wi-Fi allows the system to adapt to environments with varied infrastructure. The platform has been tested with dozens of beacons and concurrent video streams without significant performance degradation. Future deployments may incorporate distributed databases (e.g., *InfluxDB* or *TimescaleDB*) to further enhance scalability. From the deployment point of view, one of the advantages of using Docker to deploy modules is that GUIDE2FR can be deployed in a cluster of servers using *Kubernetes* and its autoscalable features like multiple replicas with Deployments and Horizontal Pod Autoscaler (HPA), load balancing with Services and the ability to dynamically add and remove cluster nodes with the cluster API, between others.

3.1. Sensor data transmission to the digital twin web portal

The system incorporates sensor data for temperature, humidity, CO_2, and total volatile organic compounds (TVOC), among other variables. These sensors are embedded within *DBEMERGENCY* beacons, as described previously, allowing deployment in critical monitoring zones. The beacons are configurable, and their data transmission rate can be adjusted based on monitoring requirements. As a result, the system provides a near real-time solution, with data displayed with only a few seconds or minutes of latency.

Data can be sourced from more than just the sensors deployed in the building. Additional inputs include alarms and processed data from other systems, as well as external *web* sources. The platform is designed to function in both training environments for first responders and public spaces, such as universities, to assess routine or emergency situations. For example, if a public building offers access to services such as classroom reservations or a Computerized Maintenance Management System (CMMS), this information can be integrated into GUIDE2FR to detect anomalies during normal operations and trigger the necessary emergency response.

Sensor data is integrated into the digital twin graphs using *Fiware*. This system activates subscription services to route information to where it is needed, facilitating long-term storage in *MySQL* and enabling time-series analysis and trend monitoring (Fig. 3).

To enable multi-step forecasting of environmental variables such as CO_2 and TVOC, we implemented a fully automated machine learning pipeline for training, hyperparameter optimization, and evaluation. The raw time series were first resampled to fixed intervals, interpolated to address missing values, and standardized using statistics derived from the training set. Lagged features (past values) and lead features (future targets) were generated to provide temporal context for supervised learning. The data were split using a forward-chaining cross-validation scheme based on rolling windows to preserve causality. Candidate models included a diverse set of regressors from the `scikit-learn` library such as Support Vector Regression (SVR), Huber Regressor, Random Forest, and Gaussian Process Regression. For each estimator, a tailored hyperparameter grid was automatically constructed by combining default values with parameter ranges suggested by a large language model, ensuring valid and model-specific search spaces. Sampling distributions (e.g., uniform, logarithmic, categorical) were selected accordingly. Each model configuration was

trained on a rolling training window (5 days), validated on the subsequent day, and evaluated across a 20-min forecasting horizon using the Mean Absolute Error (MAE). The best-performing configurations were selected based on average validation MAE across folds. Additionally, top models were combined using an adaptive ensemble strategy, where predictions were weighted inversely to their validation errors. This pipeline was tested on real-world data collected from four DBEMERGENCY beacons deployed during training exercises at Fire Station 9 (Chamartín, Madrid). It demonstrated robustness under realistic constraints and can be configured to accommodate different forecast horizons, lag depths, and sensor types.

Data acquisition and validation for model training

To ensure that the predictive models integrated in GUIDE2FR are valid in realistic emergency contexts, it was necessary to construct a dataset that addressed one of the central challenges: the high heterogeneity and unpredictability of emergency scenarios. In such contexts, environmental variables (e.g., temperature, humidity, CO_2, TVOC) evolve in ways that are tightly coupled with human intervention, spatial layout, and the dynamics of fire or ventilation. As no existing dataset offered sufficient representativeness under such conditions, we designed a data acquisition process specifically aimed at producing training data that is both ecologically valid and structurally rich. This process was carried out in collaboration with the firefighting departments of Alcorcón and Fuencarral, leveraging their operational facilities and expertise. The goal was to capture the temporal evolution of environmental variables under controlled but realistic interventions, enabling supervised training of models that can generalize to unseen but plausible emergency dynamics. The interventions were designed to indirectly perturb the environment, reproducing meaningful signal transitions without requiring actual hazardous events.

To ensure traceability and quality, the process followed seven structured steps:

1. **Initial coordination:** Firefighters provided architectural plans, inventories of standard equipment, and typical movement patterns, enabling scenario design grounded in real procedures.
2. **Deployment planning:** Based on spatial constraints and expected workflows, sensor beacons were positioned to maximize coverage of relevant zones while minimizing signal interference.
3. **Technical preparation:** All devices (beacons, routers, chargers) were validated prior to deployment. Additional materials such as labeling, signage, and procedural documentation were prepared to avoid ambiguity during execution.
4. **Operational deployment:** Sensors were physically installed in pre-selected locations, and tested in situ to verify connectivity and real-time data transmission from each zone.
5. **Protocol-driven interventions:** Firefighters executed planned sequences (e.g., entry, ventilation, movement with equipment), altering the environment in ways that reflect real operations. Each intervention was logged by participants using predefined Excel templates.
6. **Ongoing monitoring:** Throughout the campaign, communication with field staff ensured that data quality remained consistent, and technical issues were resolved without compromising the experimental integrity.
7. **Post-intervention validation:** Data was collected and pre-processed to align signal timestamps with ground-truth annotations. The resulting dataset includes time series annotated with context-specific labels (e.g., "forced ventilation started", "thermal load increase due to PPE").

By designing the data collection around firefighter workflows rather than artificial lab conditions, we ensured that the models are exposed to signal behaviors that reflect actual emergency dynamics, such as abrupt environmental changes or non-linear propagation effects. This setup provides not only a robust training base, but also a realistic benchmark to assess model performance in the kinds of contexts where their predictions will be used operationally.

3.2. Synchronized video and sensor data visualization

As described earlier, the application allows the visualization of video footage recorded by cameras synchronized with sensor data stored as time series. This synchronization is displayed in a *Grafana* dashboard using a custom *plugin* specifically developed for the platform.

Grafana[5] is an open-source data visualization and analysis platform that enables the creation of dashboards and interactive graphs. In this context, it is used to display sensor data and video in a synchronized manner. One of *Grafana's* main advantages is its support for custom *plugins* that can be integrated seamlessly into the platform.

This custom *plugin* allows the synchronization of camera-recorded video (via *OpenVidu*) with sensor data represented as time-series graphs. This solution enables interactive analysis of data and video, allowing users to navigate moments of interest in near real-time or historical information.

Each video segment is timestamped, and metadata is stored in the database by the Video Management Module. The *plugin* uses this data to synchronize RTSP video streams with time-series sensor data stored in *MySQL*. The *plugin* cross-references these timestamps to display synchronized sensor values and video playback.

Fig. 4 shows the interface of the described *plugin*. The video section features various controls to adjust playback speed or skip forward and backward quickly. Multiple videos are displayed one after another in a seamless sequence, allowing for easier and

[5] https://grafana.com/.

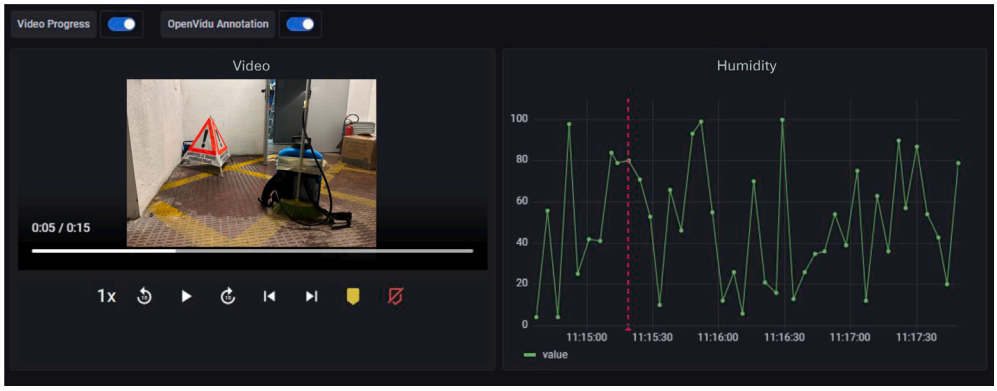


Fig. 4. *Grafana* plugin with synchronized video and time-series data.

faster review and analysis of information. Users can also tag points of interest, further facilitating easy navigation or direct access to these moments. These tags categorize information and aid in decision-making for the future Prediction System.

In emergency scenarios, this functionality allows the Advanced Command Post (PMA) chief to access footage of critical areas and, simultaneously, view environmental parameters collected by the sensors. This capability enables strategic action planning while minimizing risks to the intervention team. Historical data analysis is also possible, allowing review of recorded video alongside sensor data values at any given moment.

The *Grafana* plugin provides the following functionalities, as illustrated in Fig. 5:

- Video panel: allows to play and pause a video.
- Video connected to the TimeSeries panel: when hovering over the time series panel, the video will be updated to the time of the hovered point.
- Rewind button: allows to rewind the video to 10 s ago.
- Forward button: allows to forward the video to 10 s later.
- Video panel with annotations: allows to add a marker to the time series panel in the current time of the video.
- Skip to next annotation: allows to skip to the next annotation in the time series panel.
- Skip to previous annotation: allows to skip to the previous annotation in the time series panel.
- Annotation progress: when video is playing the annotation progress will be updated. Represented by a red line in the time series panel.

3.3. Data handling in the visualization layer

Environmental data is collected by *DBEMERGENCY* beacons. As mentioned earlier, these beacons primarily communicate via Wi-Fi, with LoRaWAN used as a fallback in case of connectivity issues during emergencies. The sensor data is first sent to a proprietary cloud API provided by the manufacturer. Periodic data retrieval is handled via *Python* scripts, which then forward the information into the *Fiware* ecosystem using NGSI interfaces. *Fiware* enables context-based data integration and distribution to the digital twin, where the information is stored in a *MySQL* database and visualized through *Grafana* dashboards and the 3D *ArcGIS*-based scene.

Sensor data processed by *Fiware* is managed through a *web* interface that provides real-time visualization of building spaces and environmental parameters.

The *web* page is developed using *Angular*, offering modularity and easy integration into existing applications as an additional component.[6] The interface is user-friendly yet powerful, featuring two primary filters (Fig. 6). One filter allows floor selection, and the other selects variables to observe. For example, the first floor of the firefighter tower is displayed with color-coded spaces, alongside graphical information on temperature, humidity, CO_2, and TVOC, shown using *Grafana* dashboards.

The *web* interface is deployed using a *Docker* image, which launches alongside the rest of the *Fiware* system. It automatically connects to the *MySQL* database to display time-series data, enabling trend analysis and real-time monitoring. The 3D model information is divided into layers: static elements like walls and staircases and dynamic elements that update with sensor data, using color codes to indicate alarm levels.

The initial filtering of digital twin layers requires creating a 3D model in software such as *Revit* or *AutoCAD* and geolocating it using *ArcGIS*. This setup enables the creation of a *web* scene and interaction with *JavaScript* tools in *ArcGIS* to filter by building floor or space attributes.

[6] https://guide2fr-cda.senialab.com/.

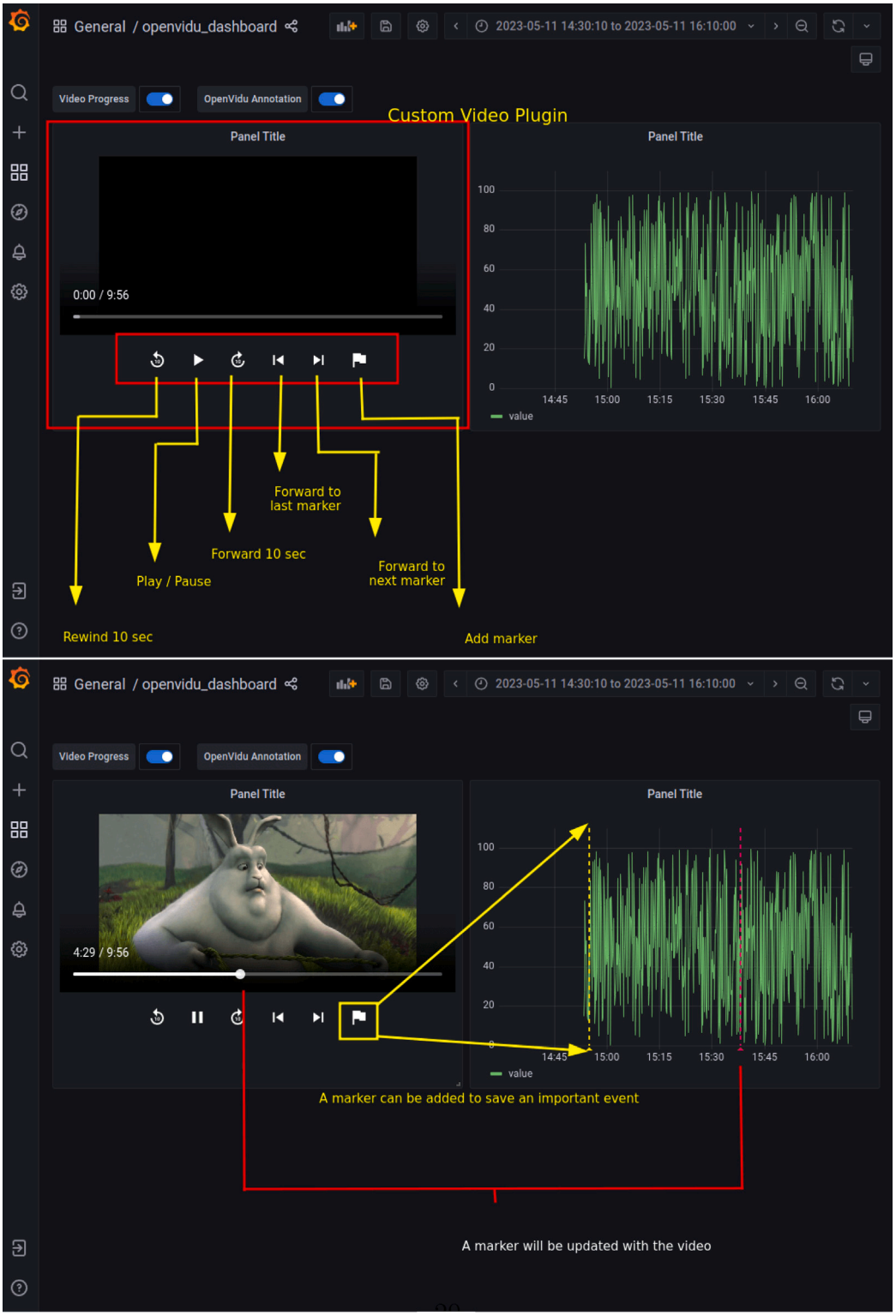


Fig. 5. *Grafana* plugin full list of functionalities.

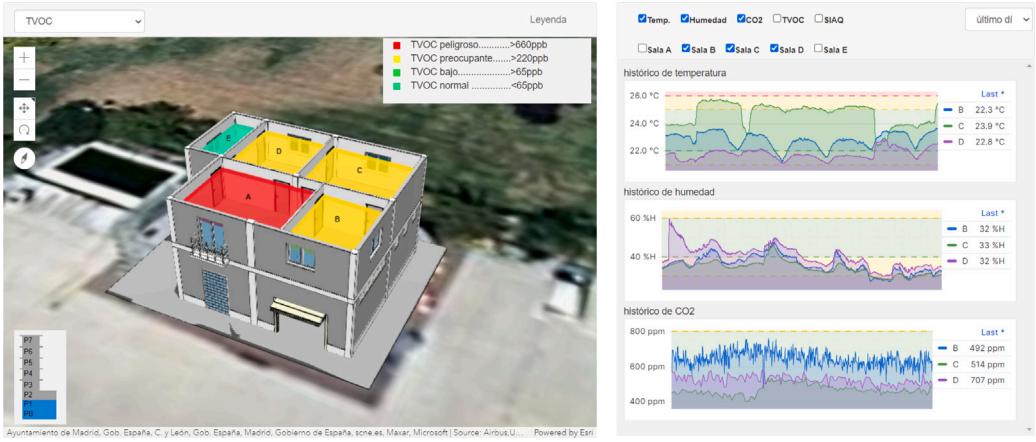


Fig. 6. Digital twin interface displaying environmental information.

Table 2
Comparison of emergency management platforms in smart buildings.

Feature	Forest Fire Detection System [6]	HelpResponder System [17]	IoT Smart Building Solutions [18]	GUIDE2FR (Proposed)
Real-time data integration	Limited to specific environmental variables	Integrates first responder safety systems but lacks comprehensive building data	Real-time data from IoT sensors, primarily for fire hazard scenarios	Comprehensive integration of environmental and structural data, visualized in a unified digital twin interface
Machine learning capabilities	Basic fire detection algorithms	Limited use of predictive analytics	No advanced predictive features	Advanced machine learning models for predicting CO_2 levels and TVOC
Scalability and flexibility	Fixed architecture, difficult to expand	Limited scalability due to specialized hardware	Moderate flexibility, focused on specific scenarios	Highly scalable and flexible architecture, easily integrates new sensors and systems
Data visualization	Basic graph-based visualization	Provides critical alerts but lacks robust visualization	Standard interfaces, not synchronized with real-time video	Advanced digital twin visualization, synchronized with real-time video and sensor data via a custom *Grafana* plugin
Open-source components	Proprietary components, limited customization	Some open-source features but generally proprietary	Mostly proprietary, with limited open-source options	Majority of components are open-source, enabling extensive customization and adaptability
Emergency management support	Focused on wildfire scenarios	Geared towards improving first responder safety	Primarily addresses fire hazards in smart buildings	Comprehensive support for diverse emergency scenarios, including training environments and public spaces

Subsequently, the layers representing variables are created using *GeoJSON*, a format that allows geolocation within *ArcGIS*. These variables are associated with sensor values or objects. *Python* scripts using the *Flask* library are employed to create the necessary *web* backend for data visualization.

Once the layers are incorporated into the scene, layer logic is designed to hide unnecessary elements and adjust the view for optimal display. Finally, rendering logic is implemented, allowing the layers to change color based on the variable being visualized and predefined alarm thresholds.

The described platform is open-source for most components, except for the *ArcGIS* software used for cartographic and building information. Therefore, replicating the platform in other environments or locations is straightforward, requiring only the adaptation of sensor and actuator data. However, obtaining map information necessitates linking with *ArcGIS* and associating each sensor and actuator with its corresponding space in the *web* portal.

4. Discussion

The evaluation of the GUIDE2FR platform has revealed several key insights, particularly in comparison to existing emergency management solutions for smart buildings. One of the main advantages of GUIDE2FR is its comprehensive integration of environmental and structural data, visualized through a digital twin interface. This capability allows intervention teams to make informed decisions in real time, improving both safety and response efficiency.

Table 2 provides a comparative analysis of GUIDE2FR against other prominent emergency management systems, including the Forest Fire Detection System [6], HelpResponder System [17], and various IoT-based Smart Building Solutions [18]. The comparison highlights several distinctive features of GUIDE2FR:

Real-time Data Integration: While other systems focus on specific variables or provide partial data integration, GUIDE2FR uniquely combines both environmental and structural data into a cohesive digital twin model. This integration enhances situational awareness and allows for a more holistic approach to emergency management.

Machine Learning Capabilities: GUIDE2FR stands out by incorporating advanced machine learning models that predict critical environmental parameters, such as CO_2 and TVOC levels. These predictive capabilities are crucial for proactive emergency response and are not commonly available in other systems.

Scalability and Flexibility: Unlike fixed or limited architectures in other platforms, GUIDE2FR's highly scalable design supports easy integration of new sensors and systems. This flexibility ensures that the platform can be adapted to various emergency scenarios and evolving requirements.

Data Visualization: The advanced digital twin visualization in GUIDE2FR, synchronized with real-time video and sensor data via a custom *Grafana* plugin, offers a superior user experience compared to basic graph-based interfaces or unsynchronized data visualizations found in other solutions.

Open-source Components: GUIDE2FR leverages mainly open-source technologies, providing a significant advantage in terms of customization and adaptability. This contrasts with the proprietary nature of many existing platforms, which often limit flexibility and increase dependency on specific vendors.

Emergency Management Support: GUIDE2FR offers comprehensive support for a wide range of emergency scenarios, including those in training environments and public spaces. This broad applicability makes it a versatile tool for first responders.

The insights derived from Table 2 underline the innovative features of GUIDE2FR and its potential to transform emergency management practices in smart buildings. The platform not only addresses current limitations but also sets a new standard for integrating real-time data, predictive analytics, and flexible architectures in urban emergency scenarios.

Despite the promising results, the proposed platform has several limitations that should be acknowledged. First, the current implementation relies on *ArcGIS* for 3D geospatial representation, which may limit portability or require licensing in certain deployment scenarios. As future work, the development of digital twin platforms based on open-source libraries and frameworks could offer more flexible and cost-effective alternatives, fostering wider adoption in diverse contexts. Second, the platform assumes the availability of a reliable wireless network infrastructure within the building, which may not always be the case during an emergency or in older facilities. Additionally, sensor calibration and performance in extreme environments (e.g., high temperature, smoke, or humidity) can impact data accuracy and model reliability. Finally, adapting the system to different building typologies or jurisdictions may require reconfiguration of sensor networks, integration with local emergency protocols, or adaptation to region-specific cloud and privacy requirements. These limitations point to important directions for future research and development to improve the platform's robustness and generalizability.

5. Conclusions

This work has demonstrated that the GUIDE2FR platform is effective for acquiring, storing, and visualizing real-world information using a digital twin. Furthermore, the platform can handle various information sources and effectively combine them to assess emergency situations.

GUIDE2FR introduces several significant innovations compared to current commercial solutions for building emergency monitoring and management. Firstly, GUIDE2FR provides comprehensive real-time data integration, combining environmental and structural parameters within a single digital twin interface. This integrated visualization capability enables faster and more accurate decision-making by intervention teams. Secondly, our platform incorporates machine learning techniques for predicting critical parameters, such as CO_2 levels and total volatile organic compounds (TVOC), an advanced feature not commonly available in commercial solutions. Additionally, GUIDE2FR's architecture is highly flexible and scalable, allowing for the easy integration of new sensors and systems, unlike commercial solutions that tend to be less adaptable. Another innovative aspect is the synchronized visualization of sensor data and real-time videos using a custom *Grafana* plugin, facilitating comprehensive emergency situation monitoring. Lastly, most GUIDE2FR components are open-source, enabling greater customization and adaptation to specific user needs—a substantial advantage over less flexible, proprietary commercial solutions. These features demonstrate that GUIDE2FR not only matches but also surpasses the capabilities of existing commercial solutions, offering a more complete and effective tool for urban emergency management.

Therefore, this work could be implemented and integrated into the university's management *web* system for preventive emergency management, opening up the possibility for research and the design of new control systems to manage more complex environments in real time.

The modularity and reliance on open-source technologies such as *Fiware*, *Grafana*, *MySQL*, and *OpenVidu* facilitate the customization and deployment of GUIDE2FR in diverse environments, ensuring interoperability with existing emergency response systems. While the current GIS visualization layer relies on *ArcGIS*, future research will explore the adoption of open-source alternatives, with the goal of increasing system portability and eliminating dependencies on proprietary software.

Future work will expand this study to integrate the digital twin with more real assets. This will allow for active response and the integration of all possible emergency assets to establish effective evacuation mechanisms. The research group is already developing related work, focusing on energy efficiency monitoring in public and private building services, given the growing importance of IoT-based monitoring in the context of energy efficiency [30]. Furthermore, to overcome current limitations related to platform dependencies, future developments will also explore the use of open-source libraries and frameworks for digital twin visualization and management.

CRediT authorship contribution statement

Marcos Delgado Álvaro: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation. **Robert Novak:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pedro Rafael Fernández Barbosa:** Writing – original draft, Visualization, Validation, Resources, Investigation. **Iván Chicano Capelo:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation. **Micael Gallego:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation. **M. Cristina Rodriguez-Sanchez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: M. Cristina Rodriguez-Sanchez reports equipment, drugs, or supplies was provided by Rey Juan Carlos University. M. Cristina Rodriguez-Sanchez reports a relationship with Rey Juan Carlos University that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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